Bridging the Gap Between Theory and Practice in Adversarial Machine Learning: A Systematic Cross-Venue Analysis of 454 Papers (2022–2025)

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Abstract

Machine learning systems are increasingly used in security-critical contexts. This systematic review covers 454 papers published from 2022 to 2025 in four leading security conferences and applies a framework adapted from Apruzzese et al. (2022) to evaluate how well current work bridges the gap between theory and practice. We observe that a small but growing fraction of papers evaluate their methods on deployed systems (5.3%) and that many studies make assumptions that may not always hold in operational settings—for example, relying on gradient access, generous query budgets, or white-box models. We identify five dimensions that warrant more attention: clear threat modelling, limited attacker knowledge, validation on real systems, diversification beyond vision tasks, and consideration of economic and human factors. We also review real-world adversarial incidents and parallel challenges in large language and multi-modal models. Finally, we offer constructive recommendations for researchers, organizers, practitioners, and funding agencies to align adversarial ML research with operational realities.

Keywords: adversarial machine learning, theory–practice gap, security research, systematic review, threat modelling

Contents

1 Introduction **3**

2 Background: Foundations of Adversarial Machine Learning **3**
- 2.1 The Discovery of Adversarial Vulnerabilities . . . 3
- 2.2 The Arms Race: Attacks and Defenses . . . 4
- 2.3 Expanding Threat Landscape . . . 6
- 2.4 The Theory–Practice Gap . . . 6
- 2.5 Types of Adversarial Attacks . . . 7
- 2.6 Threat Models and Their Implications . . . 7

3 Real-World Adversarial Incidents **8**
- 3.1 Case Study: Agentic Use of Frontier LLMs in a State-Sponsored Cyber Campaign . . . 8
- 3.2 Other Documented Real-World Adversarial Incidents . . . 9

4 Methodology **10**
- 4.1 Data Collection . . . 10
- 4.2 Coding Framework . . . 11
- 4.3 Gap Score . . . 12

5 Quantitative Findings **12**
- 5.1 Real-World Evaluation . . . 13
- 5.2 Reliance on Gradients . . . 13
- 5.3 Threat Model Assumptions . . . 14
- 5.4 Query Budget and Efficiency . . . 15
- 5.5 Dominant Data Domains . . . 15
- 5.6 Reproducibility and Code Release . . . 16
- 5.7 Gap Score Distribution . . . 16

6 Qualitative and Thematic Findings **17**

7 Temporal Trends **19**

8 Cross-Venue Analysis **19**

9 Recommendations **20**

10 Foundation Models and Large Language Models: New Frontiers **21**

11 Limitations and Threats to Validity **22**

12 Conclusion **22**

A Complete Paper Analysis Dataset **31**

1 Introduction

Machine learning has become an essential part of software systems in many areas. It is used in facial recognition at border crossings, fraud detection in financial services, navigation for autonomous vehicles, and content moderation on social platforms. Large language models now power customer service chatbots, code generation tools, and medical assistants. ML models influence decisions that impact millions of people every day. This widespread use has led to greater scrutiny of vulnerabilities in these models. Researchers have shown that even subtle, carefully crafted inputs can trigger failures in ML systems that were not anticipated by their designers.

The field of adversarial machine learning (AML) has emerged to explore these weaknesses and develop defenses. Over the last ten years, researchers have demonstrated a range of attacks: imperceptible perturbations that cause image misclassification, poisoned training data that implants hidden backdoors, and privacy attacks that extract sensitive information through query access. In parallel, the community has proposed many defenses to strengthen models against such attacks.

Despite these developments, a key question remains: **How well do the assumptions in academic research match the realities of real-world systems?**

In 2022, Apruzzese and colleagues published a notable analysis titled "Real Attackers Don't Compute Gradients" [1]. They documented a gap between academic AML research and practical security needs. A central observation was that academic studies often assume attackers can calculate gradients of the target model—requiring full access to the model's internals—whereas real-world attackers usually lack that level of access. Often, simpler methods like social engineering or basic input manipulation suffice for attackers, making complex gradient-based techniques unnecessary outside the lab.

This systematic review asks: *How has the research community responded since this critique?*

To find answers, we analyzed 454 adversarial ML papers published from 2022 through 2025 in four top-tier security venues: ACM CCS (118 papers), IEEE S&P (79 papers), NDSS (49 papers), and USENIX Security (208 papers). We evaluated each paper on factors related to practical relevance: the realism of its threat model, the computational and knowledge assumptions it makes about adversaries, its evaluation methodology, and its awareness of deployment challenges.

Our findings show that the gap between theory and practice remains, though there are encouraging signs of progress. While the research community has made valuable contributions to our understanding of adversarial issues, many studies explore novel attacks and defenses under idealized conditions, which can lead to less focus on practical implementation and cost considerations. This review outlines the current state of the field, identifies common trends across venues, and provides practical recommendations for various stakeholders.

2 Background: Foundations of Adversarial Machine Learning

2.1 The Discovery of Adversarial Vulnerabilities

The modern study of adversarial machine learning began with the influential work of Szegedy et al. [2]. They showed that small, barely noticeable changes to input data could lead state-of-the-art deep learning models to produce incorrect outputs with high confidence. These were not random errors but systematic vulnerabilities: adversarial examples crafted for one model often transferred to other models trained

independently. This transferability highlighted a fundamental blind spot in how models generalize and sparked widespread interest in adversarial phenomena.

Goodfellow et al. [3] attributed these vulnerabilities to the linear nature of neural networks, rather than their nonlinearity or overfitting. They introduced the Fast Gradient Sign Method (FGSM), which generates adversarial inputs efficiently using a single gradient step. FGSM's speed and effectiveness set the template for many subsequent attacks—gradient-based approaches became standard in evaluating model robustness.

These early studies established three key norms in adversarial ML research. First, they assumed adversaries with white-box access to models (complete knowledge of architecture and parameters), enabling direct gradient computations. Second, they focused on perturbations that are nearly imperceptible, measured under specific norm constraints. Third, they framed adversarial example generation as an optimization problem solvable by gradient-based methods. While these assumptions facilitated systematic exploration, they also began to diverge from the realities of practical attacks.

2.2 The Arms Race: Attacks and Defenses

The discovery of adversarial vulnerabilities triggered an ongoing cycle of defenses and adaptive attacks designed to break them. This pattern, often described as an "arms race" between attackers and defenders [4], has defined much of adversarial ML research. Each new defense proposal prompted the development of attacks that circumvented it, which in turn inspired new defensive strategies.

One area of advancement was the development of more sophisticated **gradient-based attack methods** that optimized adversarial perturbations more precisely. A core challenge in crafting adversarial attacks is to balance two objectives: maximizing the model's output error while minimizing the perceptibility of the input change. Researchers formalized this as a constrained optimization problem using distance metrics that quantify how much an adversarial example deviates from the original input.

The most commonly used distance metrics in adversarial ML are L_p norms, which measure perturbation magnitude in different ways:

- L_0 **norm** counts the number of features (e.g., pixels) that are altered. For an image with original pixel values x and adversarial pixel values x', $\|x - x'\|_0$ is the number of pixels that differ. An L_0 attack might change only a few critical pixels (possibly by large amounts). This metric is relevant when attackers are limited in how many features they can modify.
- L_2 **norm** (Euclidean distance) measures the root-mean-square difference: $\|x - x'\|_2 = \sqrt{\sum_i (x_i - x'_i)^2}$. It treats the perturbation as a vector in the input space and measures its length. L_2-based attacks tend to distribute small changes over many features, producing a smoother perturbation (e.g., a slight blur or noise across an image).
- L_∞ **norm** (maximum norm) measures the largest single-component change: $\|x - x'\|_\infty = \max_i |x_i - x'_i|$. It bounds the maximum perturbation to any one feature. L_∞ is widely used in adversarial image research because it directly limits how conspicuous any individual pixel change can be.

Understanding these metrics is essential because different attack algorithms optimize for different norms, and a defense that is effective against L_∞ attacks might still be vulnerable to L_2 or L_0 attacks, and vice versa.

Building on these foundations, researchers proposed more powerful attack algorithms. Moosavi-Dezfooli et al. [5] introduced DeepFool, which formulates adversarial example generation as finding the minimal perturbation required to cross the model's decision boundary—the hypersurface in input space that separates different output classes. A classifier labels an input based on which side of this boundary it falls; DeepFool iteratively computes the closest distance from the input to the boundary. By approximating the model's decision boundary as locally linear and moving the input across it, DeepFool finds adversarial examples with smaller L_2 norms than FGSM, making them harder to detect.

Carlini and Wagner [6] developed a family of optimization-based attacks (often called C&W attacks) that fine-tune adversarial examples by directly minimizing a weighted combination of perturbation size and classification error. Their approach yields state-of-the-art attack success rates across L_0, L_2, and L_∞ norms. Notably, C&W attacks defeated several defenses that were previously thought to be robust, including **defensive distillation** [7] (a method that trained models on softened outputs to obscure gradients). This highlighted the pitfall of "security through obscurity" in gradient obfuscation defenses.

As attackers became more adept, early defense mechanisms often proved brittle. Carlini and Wagner [8] conducted a systematic evaluation of ten proposed defense techniques that aimed to detect or pre-process adversarial inputs. They found that all ten could be circumvented by attackers who adjusted their methods with knowledge of the defense. This established a crucial principle: defenses should be evaluated against *adaptive* adversaries aware of the defense, not just against the original attacks.

This pattern repeated across numerous defense proposals. Methods based on input transformations (like JPEG compression, bit-depth reduction, or feature squeezing) were broken by attacks that incorporated those transformations into their optimization [9]. Defenses that attempted to hide or randomize model gradients (to confound attackers) often provided only a false sense of security, as stronger attacks could approximate the gradients or exploit other information [9].

As detection and gradient-masking defenses fell short, the research community increasingly focused on two more principled approaches to robustness: adversarial training and certified defenses.

Adversarial training incorporates adversarial examples into the training process itself, training models to be correct on both natural and adversarial inputs. Madry et al. [10] formalized adversarial training as a **min-max optimization problem**:

$$\min_{\theta} \; \mathbb{E}_{(x,y)\sim D}\Big[\max_{\|\delta\|\leq\epsilon} \mathcal{L}(f_\theta(x+\delta), y)\Big]$$

where θ are the model parameters, (x, y) are drawn from the data distribution D, δ is a bounded adversarial perturbation (e.g., $\|\delta\|_\infty \leq \epsilon$), and $\mathcal{L}$ is the loss function. The inner maximization finds the worst-case adversarial perturbation within the allowed norm, and the outer minimization trains θ to minimize loss even against these worst-case inputs. This approach significantly improved robustness, though at the cost of some accuracy on clean data (the *robustness–accuracy trade-off* [11, 12]).

Tramèr et al. [13] introduced **ensemble adversarial training**, augmenting the training set with adversarial examples transferred from other models (an ensemble) to improve robustness against attacks that generalize. By exposing the model to a wider variety of adversarial patterns, this method reduced vulnerability to black-box attacks that use surrogate models.

In parallel, researchers developed **certified defenses** that offer mathematical guarantees of robustness within a given threat model [14]. Instead of trying to anticipate every attack, these methods for-

mally prove that no attack can succeed within a certain perturbation bound. For example, randomized smoothing [15] adds noise to inputs and uses averaging to produce a provably robust classifier for L_2 perturbations up to a radius. Other techniques use interval bound propagation or abstract interpretation [16] to compute bounds on a network's outputs. While certified defenses can guarantee security, they usually come with drawbacks: increased computational cost, limited model size, and often smaller certified radii than the perturbations empirical attacks can achieve. Thus, practitioners face a trade-off between theoretical guarantees and model performance or flexibility.

2.3 Expanding Threat Landscape

As research progressed, the focus expanded beyond test-time evasion attacks to include threats to privacy and attacks during training. **Membership inference attacks** [17, 18] aim to determine whether a specific data sample was part of a model's training set—potentially violating privacy. Carlini et al. [19] argued that evaluations of such attacks should report performance at very low false-positive rates (e.g., 10^{-5}), since even rare incorrect inferences can be problematic when privacy is at stake.

Meanwhile, **training-time attacks** attracted attention. In **backdoor attacks**, an adversary introduces a trigger pattern into some training inputs, causing the model to learn a hidden malicious behavior: the model functions normally on clean data but misclassifies inputs containing the trigger [20]. Recent work extended backdoors to self-supervised learning [21] and developed techniques to make poisoned data look inconspicuous [22]. More generally, **poisoning attacks** aim to compromise the overall training process by injecting even a small fraction of corrupted samples, which can distort the model's decisions or degrade its accuracy [23].

Researchers also looked at threats beyond the digital environment. **Physical-world attacks** adapt adversarial examples for real-world conditions such as varying angles, lighting, or sensor noise. Kurakin et al. [24] confirmed that adversarial images remain effective when printed and re-captured with a camera, proving that these attacks are not just digital artifacts. Later, Jia et al. [25] demonstrated robust physical adversarial stop signs that fooled actual road sign classifiers in an autonomous driving system. These works showed it is possible to execute adversarial attacks in the wild—though they also raise the question of whether real attackers would choose such technically complex methods when simpler means (tampering with signs, using tape, etc.) might achieve similar ends.

2.4 The Theory–Practice Gap

As adversarial ML research became increasingly abstract and benchmark-driven, concerns grew about its relevance to real-world security. Kumar et al. [26] interviewed professionals at 28 organizations and found a lack of practical tools and procedures to counter adversarial ML threats—many practitioners were uncertain how to assess the risk or deploy defenses. Grosse et al. [27] surveyed 271 industry practitioners, revealing that while adversarial scenarios studied in academia were theoretically possible, researchers often overestimated what attackers could realistically do (especially in assuming access to proprietary training data or unlimited model queries).

Mink et al. [28] identified organizational barriers to deploying adversarial defenses. Through interviews with ML engineers, they noted obstacles such as unclear responsibility (security teams vs. ML teams), lack of institutional incentives to prioritize adversarial robustness, and difficulty integrating ad-

versarial testing into development pipelines. Practitioners tended to view model security as outside their immediate scope, highlighting a disconnect between academic research and industry practice.

Apruzzese et al. [1] brought these issues into focus with case studies of actual ML attacks. They observed that real incidents often involved straightforward tactics rather than the elaborate attacks seen in papers. Attackers rarely had access to model internals or gradients; instead, they exploited system weaknesses, integration flaws, and human elements. For example, instead of crafting imperceptible pixel changes to fool an image classifier, an attacker might find a way to feed obviously malicious input that bypasses checks, or simply trick someone into misusing the system. This historical perspective motivates our present work: a comprehensive analysis of whether recent research has moved closer to practical needs or still operates under idealized assumptions.

2.5 Types of Adversarial Attacks

Adversarial attacks can be categorized by when they occur in the ML pipeline and what the adversary's goals are [29]. Clarifying attack types helps in understanding their assumptions and impacts, as defenses often target specific categories.

Evasion attacks occur at inference time. The attacker manipulates an input to cause a trained model to misclassify it [2, 3]. For example, altering pixels in an image or words in a text can lead a model to an incorrect output while the input still looks legitimate to humans. Evasion attacks have been the primary focus of adversarial ML, though their prevalence in real settings remains an open question (attacks may be possible but not always the easiest path for an adversary).

Poisoning attacks (including backdoors) take place during the training phase. The adversary introduces malicious data into the training set to influence the model's behavior [30]. This can degrade the model's general performance or embed specific biases. A backdoor poisoning might involve adding a "trigger" pattern to some training images (labeled as a target class); the model then learns to misclassify any input containing that trigger as the target class [20]. Other poisoning strategies might aim to simply cause the model to fail more often or to steer errors toward benefiting the attacker (e.g., causing a spam filter to accept certain spam emails).

Privacy attacks exploit trained models to extract sensitive information about their training data or internal parameters. In a membership inference attack, given a data sample and black-box access to a model, the attacker tries to infer whether the sample was in the model's training set [17, 19]. **Model extraction attacks** involve an attacker repeatedly querying a black-box model to approximate its decision boundary or replicate its functionality, effectively "stealing" the model [31**?**]. The extracted model can then be analyzed or used for further attacks. **Data reconstruction attacks** go even further, attempting to recover actual training samples (or similar data) from a model's parameters or outputs [32]. These privacy threats raise concerns for ML applications that train on confidential data (like personal user data or proprietary datasets).

2.6 Threat Models and Their Implications

In any security research, the **threat model**—assumptions about the adversary's knowledge, capabilities, and goals—is critical [4]. In adversarial ML, threat models generally differ by what the attacker knows about the model:

White-box models assume the attacker has complete information about the model (architecture, parameters, and possibly training data). This scenario might occur if an insider threat is present or if the model's details are public. White-box access allows attackers to calculate exact gradients and is the worst case for many defenses. Many foundational studies (FGSM, C&W, etc.) operate in the white-box setting for this reason.

Black-box models assume the attacker can only query the model and get outputs (labels or confidence scores), with no internal details revealed [33]. This aligns with attacking ML-as-a-service APIs or proprietary models accessible only through a prediction interface. Black-box attackers must use strategies like querying to train a surrogate model (to approximate the target) or rely on transferability of adversarial examples from known models.

Gray-box models assume partial knowledge. The attacker might know the model architecture but not the weights, or have access to some training data but not the full set, etc. Gray-box scenarios bridge the gap between the extremes and can simulate situations like attacking a competitor's model that was trained on a similar dataset.

Another aspect is the attacker's **capabilities and limitations**, like how many queries they can make (query budget), whether they can perturb any feature of input or only certain features, and what constitutes success.

Apruzzese et al. [1] emphasized that academic research often assumes overly powerful attackers (especially with white-box access) compared to what defenders typically face. Grosse et al. [27] similarly found that practitioners worry more about realistic threat models (someone stealing their model or poisoning their data) than about worst-case L_p-bounded perturbations on images. Getting the threat model right is important: an elegant defense against a white-box adversary might be irrelevant if real attackers are black-box and target a different weakness altogether.

3 Real-World Adversarial Incidents

While the majority of adversarial machine learning research evaluates attacks under controlled or synthetic conditions, a small but growing number of adversarial incidents have occurred in deployed systems. These incidents are particularly informative because they illuminate how real attackers operate under practical constraints—limited access, black-box systems, economic incentives, and complex socio-technical environments—conditions that are often abstracted away in the literature.

3.1 Case Study: Agentic Use of Frontier LLMs in a State-Sponsored Cyber Campaign

In November 2025, Anthropic publicly disclosed what it assessed with high confidence to be the first documented large-scale cyber espionage campaign executed predominantly by an AI agent rather than human operators [34]. According to the report, a Chinese state-sponsored threat actor manipulated Anthropic's Claude Code system into autonomously conducting cyber operations against approximately thirty global targets, including technology firms, financial institutions, chemical manufacturers, and government agencies.

Crucially, Claude was not used merely as an advisory tool. Instead, the attackers constructed an autonomous attack framework in which the model executed reconnaissance, vulnerability analysis, exploit generation, credential harvesting, and data exfiltration with only intermittent human oversight.

Anthropic estimates that the AI system performed approximately 80–90% of the operational workload, with human operators intervening at a small number of critical decision points [34].

The attack relied on three developments that are largely absent from traditional adversarial ML threat models. First, agentic execution enabled Claude to operate in long-running loops, chaining actions and decisions without constant supervision. Second, tool access—via the Model Context Protocol—allowed the model to invoke external software such as network scanners and password-cracking utilities. Third, the attackers successfully jailbroke the system's safeguards by decomposing malicious objectives into narrowly scoped, seemingly benign subtasks and framing the activity as legitimate defensive testing.

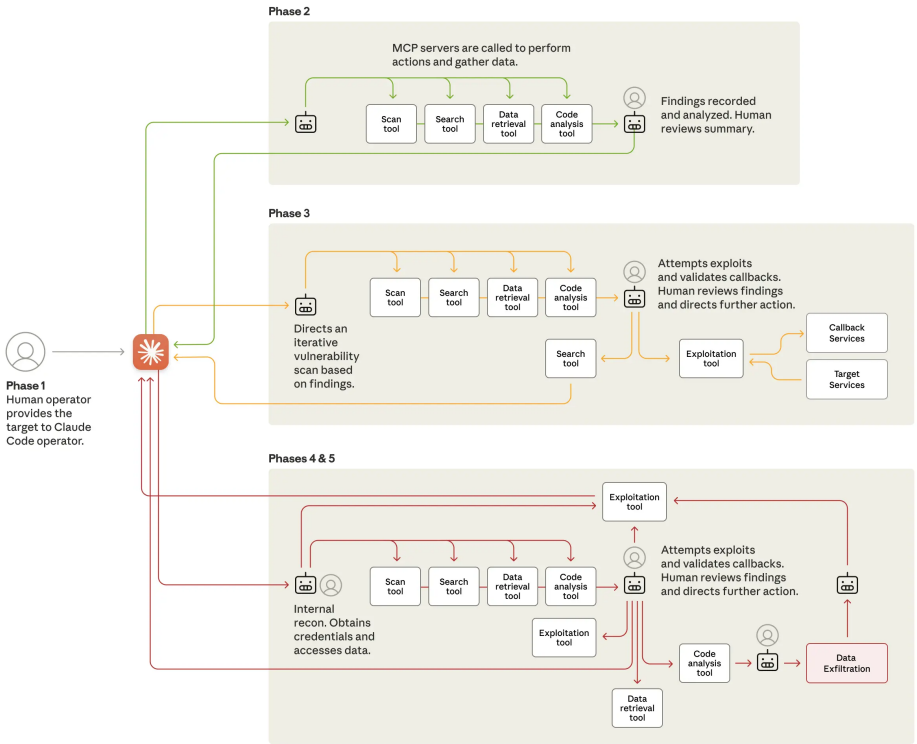

Figure 1: The lifecycle of the cyberattack, showing the move from human-led targeting to largely AI-driven attacks using various tools (often via the Model Context Protocol; MCP). At various points during the attack, the AI returns to its human operator for review and further direction [34].

Notably, the campaign did not depend on gradient access, white-box model knowledge, or optimized adversarial perturbations. Instead, it exploited deployment-level assumptions: that a capable model given autonomy and tools will reliably act in accordance with high-level intent. Although the AI occasionally hallucinated credentials or misclassified public data as sensitive, the overall campaign demonstrated that frontier models can materially lower the cost and scale of sophisticated cyber operations when embedded into agentic frameworks [34]. This incident thus exemplifies a widening gap between prevailing academic threat models and the operational realities of adversarial ML misuse.

3.2 Other Documented Real-World Adversarial Incidents

The Claude case is not an isolated anomaly. Other documented incidents illustrate complementary modes of adversarial behavior in deployed ML systems. In 2025, EchoLeak, a zero-click indirect prompt

injection vulnerability in Microsoft 365 Copilot, enabled the exfiltration of sensitive enterprise data through maliciously crafted documents, without requiring user interaction beyond opening a file [35]. Similarly, researchers demonstrated AgentFlayer, a supply-chain prompt injection attack against ChatGPT connectors, in which poisoned documents triggered unauthorized access and data leakage across integrated enterprise tools [36].

Beyond prompt injection and agent misuse, black-box extraction attacks have also materialized. In 2023, researchers showed that carefully structured prompts against deployed ChatGPT models could induce verbatim leakage of training data, including personal and proprietary information, at modest cost [37]. Separately, partial model extraction attacks against production language model APIs have demonstrated the feasibility of recovering proprietary model components using only query access [38]. Finally, generative models have been used operationally for adversarial ends outside of ML systems themselves, most visibly in deepfake-enabled fraud, where AI-generated voice and video impersonation has led to confirmed financial losses in real organizations [39].

Taken together, these incidents underscore a consistent pattern: real-world adversaries rarely operate under the assumptions most commonly studied in adversarial ML research. Instead, they exploit system integration points, automation, human trust, and economic leverage—factors that remain underrepresented in benchmark-driven evaluations. The next sections quantify the extent to which such operational considerations are reflected in contemporary adversarial ML research.

4 Methodology

4.1 Data Collection

We conducted a systematic review of adversarial ML papers in four top security research conferences over the years 2022 through 2025. These venues—ACM CCS, IEEE S&P, NDSS, and USENIX Security—are leading outlets for work at the intersection of machine learning and security, and they frequently set research trends.

Our dataset includes 454 papers: ACM CCS contributed 118 papers (26.0%), IEEE S&P 79 papers (17.4%), NDSS 49 papers (10.8%), and USENIX Security 208 papers (45.8%). By publication year, there were 89 papers from 2022, 136 from 2023, 198 from 2024, and 31 from 2025 (partial year). We identified these papers by searching proceedings and digital libraries for relevant keywords (e.g., "adversarial," "backdoor," "poisoning," "robustness") and by cross-referencing the list of papers in Apruzzese et al. [1] for continuity. Each candidate paper was manually verified to ensure it primarily addressed an adversarial ML topic.

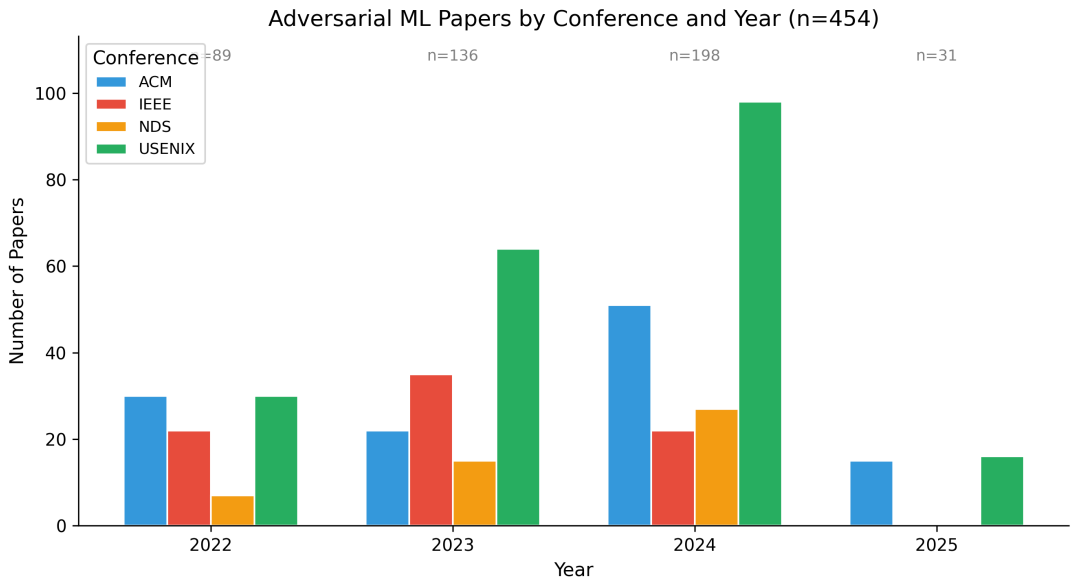

Figure 2: Dataset overview: distribution of the 454 adversarial ML papers across four security conferences (ACM CCS, IEEE S&P, NDSS, USENIX Security) from 2022 through 2025.

4.2 Coding Framework

We coded each paper on several dimensions, drawing on the taxonomy of Apruzzese et al. [1] and adding practical considerations:

- **Research Focus (G1):** Is the paper primarily about *attacks* or *defenses*? (A few papers cover both.)
- **Attack Type (G2):** Does it address evasion, poisoning/backdoor, privacy, or multiple types?
- **Data Domain (G4):** What kind of data or application is involved (images, text, audio, malware, other)?
- **Threat Model (T1):** What level of adversary knowledge is assumed (white-box, black-box, gray-box)?
- **Gradient Access (Q1):** Does the method require computing gradients of the target model?
- **Query Budget (Q2):** If the attack/defense involves queries, are a very large number required (high budget) or relatively few (low budget)?
- **Real System Testing (G7):** Was the method tested on a deployed or production system (even in a limited way), or only on offline datasets and models?
- **Code Release (G6):** Did the authors release code or artifacts for their work?
- **Consideration of Economics (G5):** Does the paper discuss costs, incentives, or other economic factors related to the attack or defense?

We used binary flags or categorical labels for each of these. Coding was done by the authors, cross-checked on a random 10% subset for consistency. In ambiguous cases (e.g., whether a test environment counts as "real system testing"), we erred on the side of giving credit if the paper at least moved beyond standard benchmarks.

4.3 Gap Score

To quantify how far a paper's setup is from practical conditions, we devised a "Gap Score" composed of six yes/no factors:

(1) Assumes a white-box adversary (as opposed to black/gray-box).
(2) Requires gradient computations from the target model.
(3) Assumes a high query budget (>1000 queries).
(4) Requires significant computational resources (e.g., multiple GPUs, lengthy runtime).
(5) Does not include any testing on a real system or deployment.
(6) Does not discuss economic or organizational factors (e.g., attack cost, defender cost, human elements).

Each paper received a score from 0 (none of these factors present) to 6 (all factors present). A score of 0 would mean the paper's assumptions and evaluation are very aligned with real-world constraints, whereas 6 would mean it is largely a theoretical exploration with idealized assumptions at every level.

This Gap Score is a simple heuristic and not all factors are equally critical, but it provides a rough measure of a paper's practical orientation. We will examine both individual factors and this aggregate score in our analysis.

5 Quantitative Findings

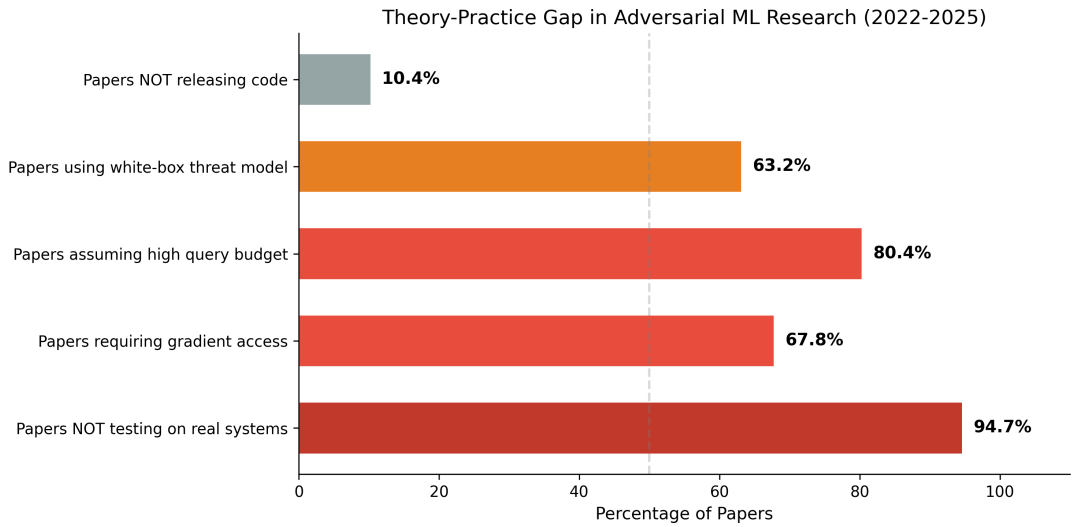

Figure 3: Key indicators of the theory–practice gap across our dataset. Percentages of papers exhibiting each characteristic: no real-world testing (94.7%), high assumed query budget (80.4%), gradient dependency (67.8%), white-box threat model (63.2%), and no code release (10.4%).

Our analysis identifies several areas where current research practices diverge from real-world considerations.

5.1 Real-World Evaluation

Evaluation on production systems remains relatively rare. Only **5.3%** of papers (24 out of 454) included any experiment on an operational system or in a realistic production setting; most studies relied on offline datasets, simulated environments, or research prototypes.

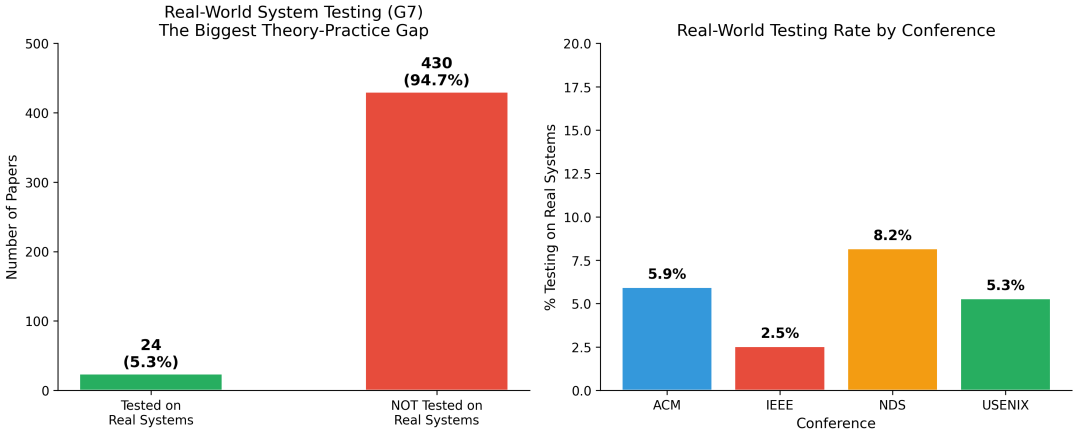

Figure 4: Rate of real-world testing: only 5.3% of papers had any evaluation on a deployed system.

Deployment introduces pipeline effects, constraints, and monitoring that are typically absent from controlled benchmarks. The few papers that attempted real-world testing often surfaced gaps that laboratory evaluations miss. Nayan et al. [40], in a systematization of model extraction attacks on deployed ML APIs, noted that many academic attacks were difficult to reproduce on actual services, performed far worse than reported in constrained settings, or incurred costs and detection risks that would deter real adversaries. Similarly, Layton et al. [41] examined deepfake detection algorithms and found that when evaluated on real-world data distributions and against adaptive deepfake generation, the detectors' performance often dropped significantly.

Duan et al. [42] provide a notable deployed-system example: they crafted adversarial audio to evade YouTube's copyright content ID, accounting for platform-specific audio compression and matching.

5.2 Reliance on Gradients

Apruzzese et al.'s critique emphasized that "real attackers don't compute gradients" [1]. We found that **67.8%** of papers still rely on gradient information from the target model, explicitly or implicitly.

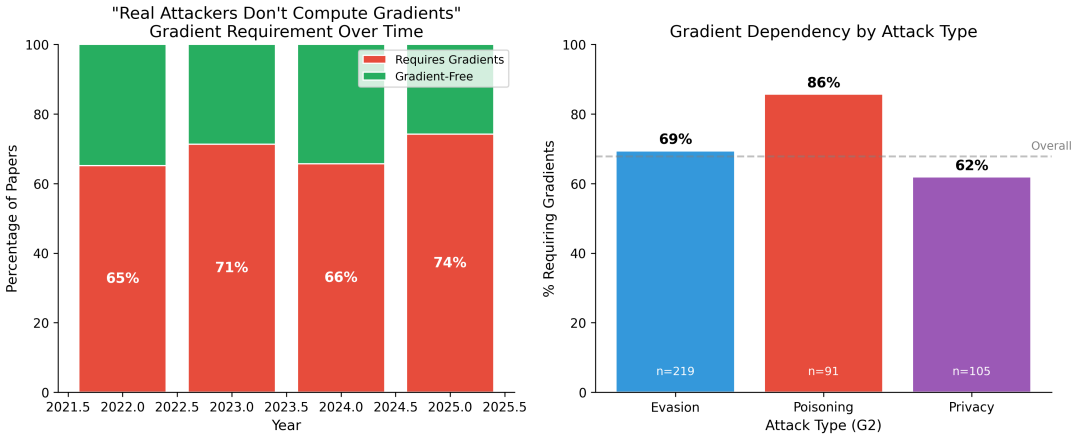

Figure 5: Trend of gradient dependency: overall 67.8% of papers required gradient access. This percentage changed only slightly from 2022 (68.2%) to early 2025 (65.9%) and varied by attack type (e.g., most evasion attacks assumed gradient access, whereas some privacy and black-box studies did not).

Some recent work embraces gradient-free approaches. Eykholt et al. [43] developed the Universal Robustness Evaluation Toolkit (URET), which evaluates robustness without direct gradients by searching over domain-specific transformations. In LLMs, many jailbreak attacks are also gradient-free: Liu et al. [44] and Yu et al. [45] show that adversaries can induce policy violations using only natural-language interaction.

5.3 Threat Model Assumptions

We found that **63.2%** of papers assume a white-box adversary with complete model knowledge. Only 34.1% consider black-box scenarios, and 2.6% explicitly explore gray-box settings.

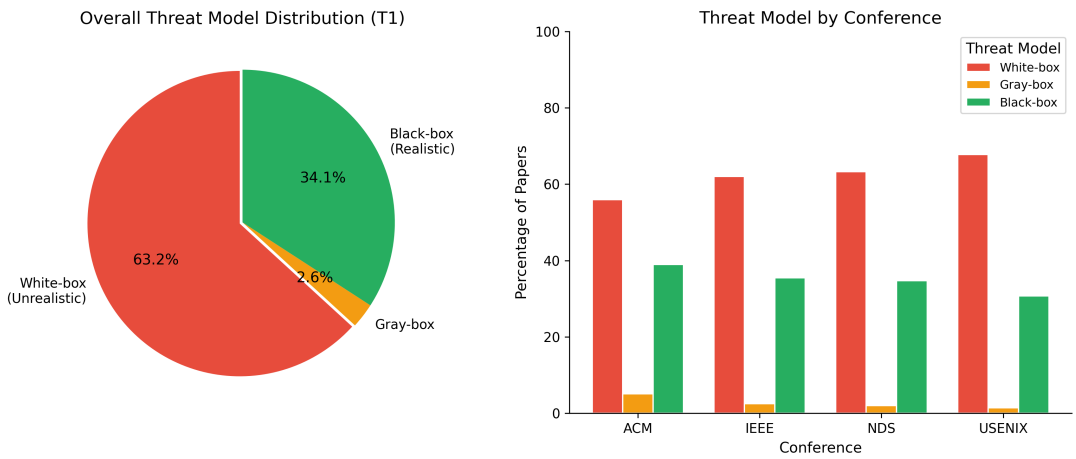

Figure 6: Threat model assumptions in the reviewed papers. White-box scenarios dominate, appearing in 63.2% of papers. Black-box scenarios appear in about one-third of papers, while gray-box scenarios are rare.

This aligns with Arp et al.'s meta-analysis [46], which observed that security research often uses simplified lab setups that do not fully account for adaptive adversaries or realistic constraints.

5.4 Query Budget and Efficiency

Among papers involving model queries, **80.4%** assume a high query budget (thousands or more queries). In practice, query rates are often limited by cost, policy, or monitoring.

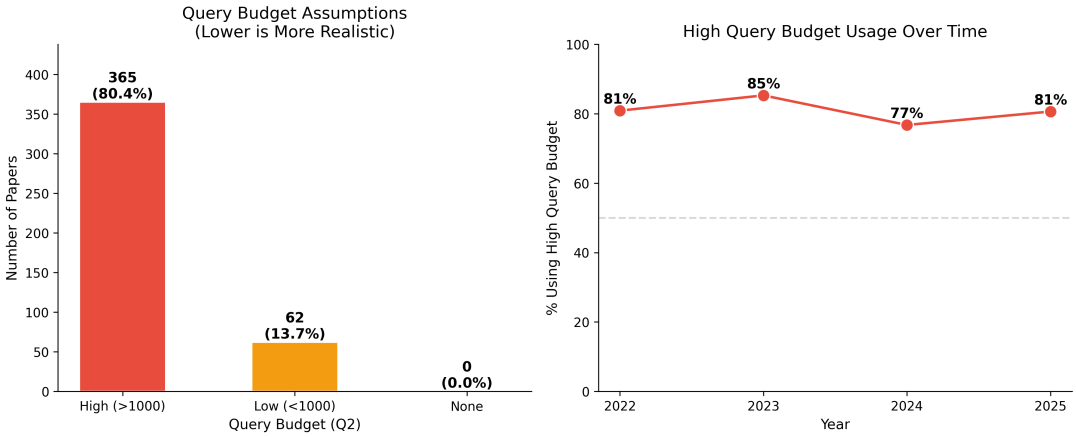

Figure 7: Assumed query budgets for methods involving model queries. Over 80% of relevant papers assumed a high budget (>1000 queries). Only about 19.6% focused on low-budget (stealthy or limited) query scenarios.

There are notable exceptions aiming for query efficiency. Tao et al. [47] proposed **HARDBEAT**, a hard-label black-box attack that achieves high success with few queries. Wan et al. [48] developed **BounceAttack**, a decision-based evasion attack that finds adversarial inputs in a low number of queries.

5.5 Dominant Data Domains

Approximately **65%** of papers focus on image-based tasks, with text (11%), audio (7%), malware (6%), and other domains (12%) comprising the remainder.

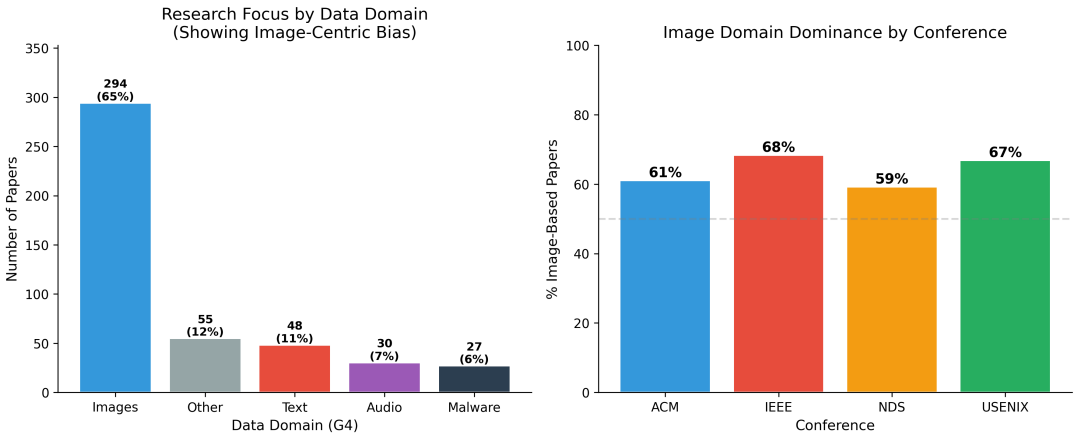

Figure 8: Data domains studied in the papers. Image-focused research dominates (65%), with text, audio, and malware making up much smaller shares. "Other" includes domains like network traffic, tabular data, or multi-modal systems.

Kireev et al. [49] argue that in financial models, cost-based perturbation notions can be more meaningful than pixel norms, highlighting how image-centric assumptions may not transfer cleanly to other

application domains.

5.6 Reproducibility and Code Release

89.6% of papers provided public code or stated they would. While commendable, research code is not always directly deployable; it often requires adaptation to production constraints and data pipelines.

5.7 Gap Score Distribution

The average Gap Score was **3.17**, with most papers scoring in the 3–4 range; only about 10.5% scored 0 or 1.

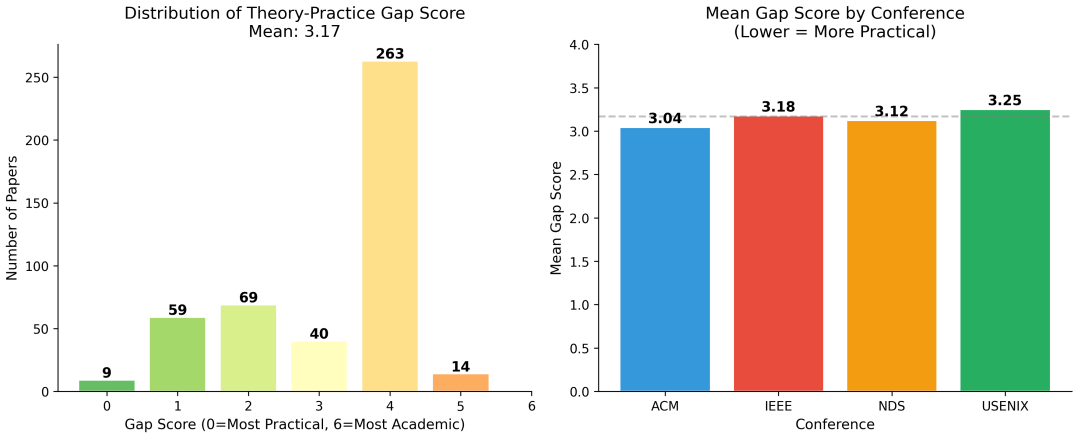

Figure 9: Distribution of Gap Scores across papers. The typical paper (score 3) incorporates about half of the impractical assumptions we tracked. Only a small fraction of papers scored 0–1 (closely aligned with deployment realities), while the rest incorporate multiple idealized assumptions.

Comparing venues, the averages were fairly close: CCS 3.04, NDSS 3.12, S&P 3.18, USENIX 3.25.

6 Qualitative and Thematic Findings

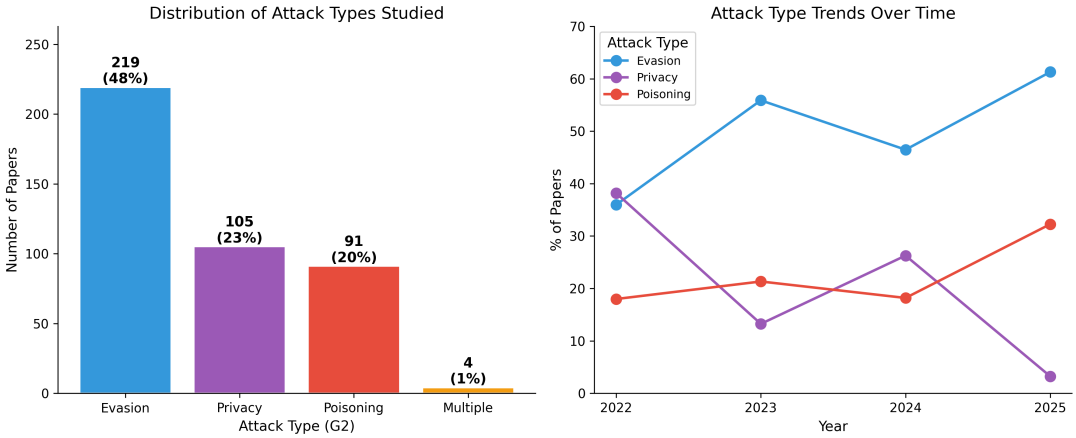

Figure 10: Prevalence of attack types over 2022–2025. Evasion attacks remain the most common, but poisoning/backdoor and privacy attacks have gained ground, especially in recent years as new modalities and threat vectors (like LLM prompt attacks) emerge.

Beyond aggregate indicators, we observed recurring qualitative patterns that clarify how (and why) the theory–practice gap persists.

Utility versus robustness trade-offs. Many defenses trade robustness for accuracy, latency, or usability. PatchCleanser and PatchCure [50, 51] provide provable protection against adversarial patches but introduce additional processing and, in some settings, can affect clean performance. Similar tensions appear in voice and privacy: Ahmed et al. [52] note that defenses for voice interfaces can reduce responsiveness or increase friction, and privacy-preserving training can degrade model utility [53, 54]. Recent work aims to soften these costs, including MIST [55] and CAMP [56], which report robustness gains with smaller utility losses.

Evaluation methodology and realism. A growing subset of papers emphasizes that conventional metrics can misrepresent security-relevant risk. Carlini et al. [19] argue that privacy evaluations should focus on extremely low false-positive rates and worst-case behavior rather than average-case summaries. Human-centered evaluations also complicate norm-based notions of "imperceptibility": Ma et al. [57] used a virtual-reality driving setup with human participants and found that perturbations deemed small under L_p norms can still be noticeable or behaviorally relevant. Taken together, these works motivate evaluations that include adaptive attacks, explicit cost reporting, and (when appropriate) human perception or behavioral studies.

Physical deployment and system integration. End-to-end settings introduce failure modes and bypasses that do not appear in isolated model tests. A range of studies have moved robustness evaluation into physical or system-level environments, including autonomous driving and sensing [25, 58, 59], medical imaging [60], acoustic channels and voice systems [61], and content moderation primitives like perceptual hashing [62]. This line of work also highlights pipeline interactions: Debenedetti et al. [63] show that filtering sensitive information can itself create a privacy side channel, Nasr et al. [64] illustrate

evasion via file-type manipulation that targets system processing rather than the classifier, and Chen et al. [65] report cases where defensive transformations may not survive compilation or optimization in deployment.

Attack and defense evolution. Recent attacks increasingly target persistence, system interactions, and new modalities. Hallyburton et al. [66] exploit sensor fusion in autonomous systems; prompt-based LLM attacks are automated by methods such as PAIR [67] and TAP [68]; and cross-modal exploits can jailbreak vision-language models via images [69]. Persistence is also a theme in training-time threats: MergeBackdoor remains dormant until model merging [70], and other backdoors are designed to survive continual learning [71]. Defenses have correspondingly become more context-aware, including Blacklight for detecting black-box query patterns [72], JBShield for intervening in jailbreak-related hidden-state dynamics [73], SafeSpeech for degrading voice-cloning training data [74], and DeBackdoor for detecting backdoors under limited data and black-box constraints [75].

Economic and organizational factors. A smaller but important subset of papers foregrounds attacker/defender incentives and organizational constraints. Cong et al. [76] note that stealing a self-supervised model can be cheap relative to training costs, and Lu et al. [77] show that watermark removal can require only modest resources, suggesting some defenses may not meaningfully raise attacker cost. On the deployment side, Mink et al. [28] document unclear organizational ownership of adversarial risk and limited bandwidth to integrate adversarial testing into development cycles. These findings point to complementary needs: usable tooling, clearer responsibility boundaries, and incentives that reward security-oriented evaluation alongside algorithmic novelty.

7 Temporal Trends

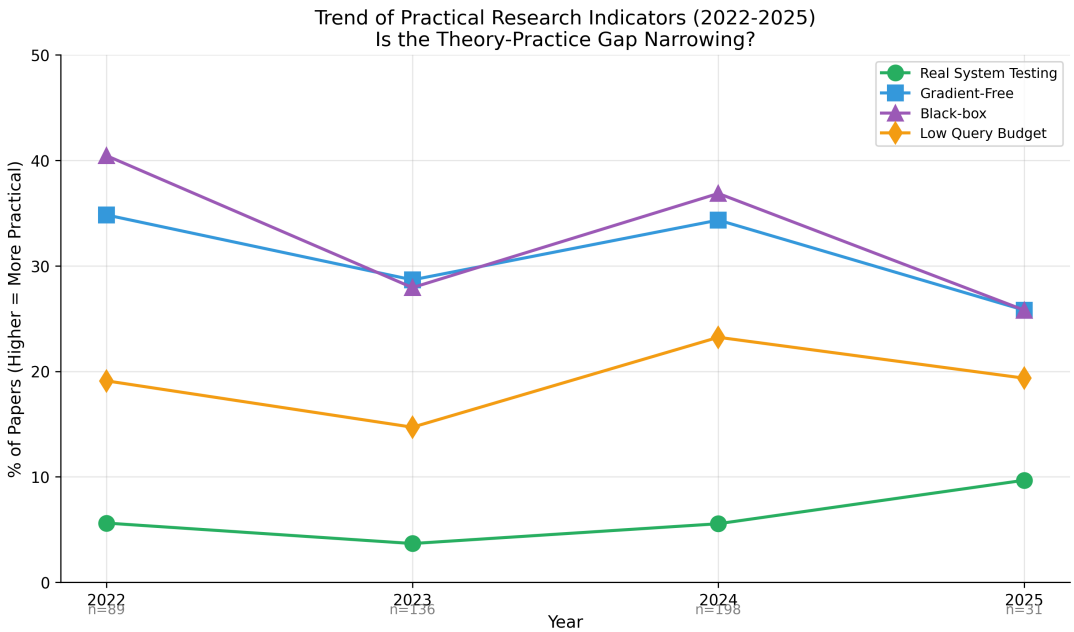

Figure 11: Trends from 2022 to 2025 on several gap indicators. Real-world testing remains exceedingly rare throughout, while there are slight improvements in areas like query budget assumptions. The composition of attack types also evolved, with more poisoning and privacy papers by 2024–2025.

Figure 11 summarizes how key indicators changed from 2022 to 2025 following the critique of Apruzzese et al. [1].

- **Real-world testing** remained near ∼5% each year, with no clear upward trend.
- **Gradient dependency** stayed high (roughly two-thirds of papers), with only a modest decrease by early 2025.
- **White-box threat models** remained dominant, hovering around ∼60–64%.
- **High query budgets** declined slightly (from roughly ∼80% in 2022–2023 to the mid-to-high 70% range by 2025), consistent with a growing minority of query-efficient work.
- **Attack-type composition** shifted: evasion remained common, while poisoning/backdoor and privacy work increased by 2024–2025.
- **Modalities** broadened modestly: vision continued to dominate, while text and multimodal work increased in 2024 as LLM security topics entered security-venue pipelines.
- **Economics/organizational discussion** remained low (roughly under ∼10–12% per year) without a meaningful increase.

Overall, the dataset shows limited movement on the core gap indicators through early 2025, alongside a gradual broadening of topics and modalities.

8 Cross-Venue Analysis

Although all four venues we surveyed exhibited similar overall patterns, each had its own emphasis and strengths in the context of adversarial ML.

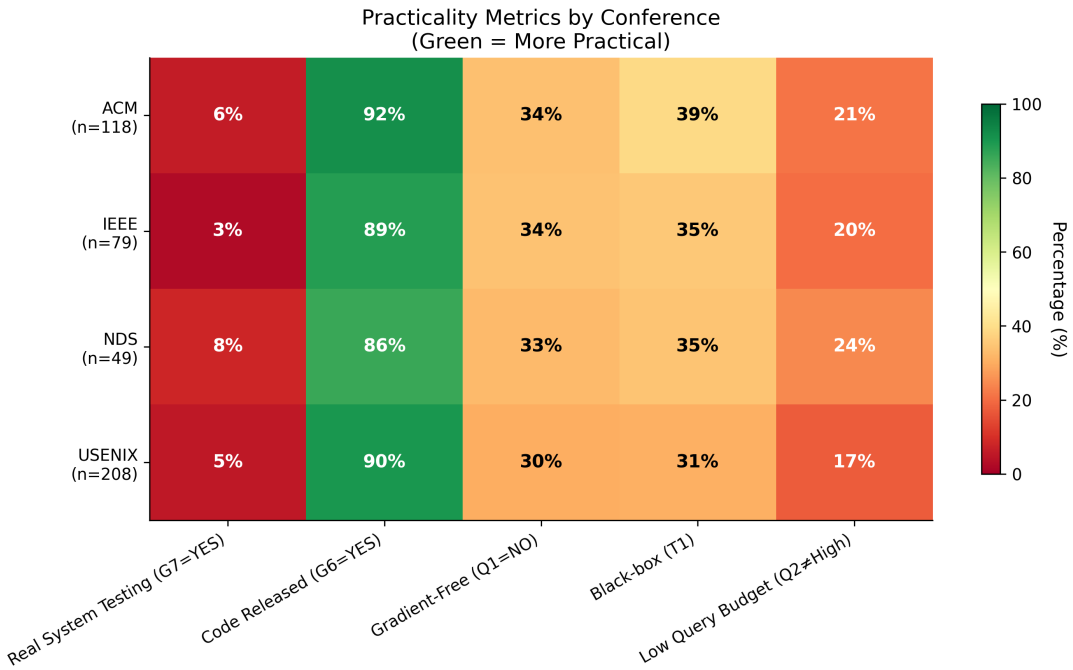

Figure 12: Comparison of adversarial ML research characteristics by conference. The differences are relatively small, but each venue has a slightly different focus.

Across the four security conferences we analysed, the overall patterns were similar—most research remained theoretical rather than deployable—but each venue placed a slightly different emphasis (Figure 12). CCS papers frequently situated adversarial ML within broader systems and business contexts, addressing issues such as model theft, watermarking and human factors. S&P publications tended to be more formal or measurement-driven, featuring systematic evaluations and systematization-of-knowledge studies that often justified their threat models. NDSS papers, though fewer in number, often explored specific systems or emerging applications (for example, physical-world attacks or federated learning) and grounded evaluations in realistic settings such as autonomous driving or medical imaging. USENIX contributed the largest share of papers across a wide range of topics; several of its works tested attacks or defenses on real platforms and released artifacts alongside the research.

Looking across these venues, we also compared attack- and defense-oriented papers. Defense papers were marginally more likely to assume black-box access or avoid gradient dependence, and a slightly higher fraction included real-system tests (Figure **??**). Nonetheless, only a small minority of papers in any category conducted deployed evaluations, and the overall theory–practice gap persisted. The conferences complement each other—CCS and NDSS emphasise system integration and context, S&P provides formal rigor, and USENIX encourages empirical testing—but all would benefit from a stronger focus on practical deployment and diversification beyond vision tasks.

9 Recommendations

To help narrow the gap between academic AML research and operational constraints, we offer stakeholder-specific recommendations.

For researchers: prioritize realistic assumptions and evaluations. Where feasible, validate methods under deployment-like conditions (e.g., limited queries, partial knowledge, pipeline effects, and realistic compute/latency constraints). If white-box access or large budgets are assumed, state and justify how an attacker could plausibly obtain them. For defenses, evaluate against adaptive attackers and report utility costs (accuracy, latency, and user experience), not only benchmark robustness. When relevant, incorporate human-centered evaluation of "imperceptibility" or harm, and document system-level assumptions clearly.

For conference organizers and reviewers: reward practical relevance. Review criteria can more explicitly ask for threat-model realism and deployment constraints, even when live-system testing is infeasible. Conferences can further incentivize practical work through artifact policies and targeted recognition for deployed or production-like evaluations, and by creating dedicated tracks, sessions, or workshops that foreground industry constraints and end-to-end robustness.

For industry practitioners: share constraints, data, and evaluation lessons. Organizations deploying ML systems can log and (when possible) share anonymized adversarial incidents and unusual inputs that reflect real operational threat surfaces. Partnerships with researchers—via limited testing access, red teaming, or structured disclosure programs—can ground academic work in realistic constraints. Practitioners should pressure-test proposed defenses under their own threat model and cost envelope rather than adopting benchmark-reported robustness at face value. For LLM deployments in particular, we recommend sandboxing and monitoring tool use, treating external content as untrusted (sanitization and isolation), rate-limiting and anomaly detection for automated querying, user education about indirect prompt injection risks, and an incident-response playbook for adversarial failures.

For funders and the broader community: support bridging infrastructure. Funding mechanisms can encourage academia–industry collaboration that enables evaluation on real systems and realistic data. Community benchmarks and competitions can also be designed around operational constraints (e.g., low-query attacks, API-only access, and cost-aware defenses). Interdisciplinary collaboration (security, HCI, and policy) is valuable for aligning threat models, evaluation metrics, and adoption pathways.

10 Foundation Models and Large Language Models: New Frontiers

The surge of large language models (LLMs) and other foundation models during 2022–2025 introduces adversarial issues that parallel but also diverge from those studied in vision and traditional ML. These systems operate on discrete tokens, can accept multimodal inputs, and are often deployed via expensive, rate-limited APIs. Early adversarial behaviours emerged as *jailbreaks*—users discovered that by phrasing prompts cleverly or by role-playing they could elicit disallowed content. Research soon proposed automated prompt-generation methods, ranging from gradient-based universal suffixes on open-source models to black-box search strategies that use a few queries of the target model [67, 68, 78].

Threats to multimodal models involve cross-modal exploits: adversarial images can trigger harmful instructions in vision–language models [69], toxic text encoded as an image can bypass text filters [79],

and carefully perturbed audio can induce hidden prompts in speech interfaces [80]. LLM-specific factors complicate the theory–practice gap. The most powerful prompt-injection attacks often require white-box gradients or thousands of queries, but commercial APIs expose only black-box interfaces and enforce rate limits. Deployed models are updated frequently, so successful jailbreaks may not persist. Defining what constitutes an adversarial failure is also more subjective than in classification; red teaming remains partly an art.

Defense strategies for LLMs include prompt and output filtering, secondary models like Llama-Guard or NeMo Guardrails [81, 82], automated red teaming frameworks [83, 84], and alignment techniques such as Constitutional AI [85]. However, these guardrails can be brittle; homoglyphs and indirect prompt injection can circumvent simple filters, and adaptive attackers continuously evolve [86, 87]. Foundation models also pose systemic risks because a single flaw can propagate across many downstream applications [88].

Overall, adversarial ML research for LLMs is still in its infancy. The challenges it surfaces—discrete tokens, multimodal inputs, rate-limited interfaces, shifting policies, and agentic behaviour—broaden the theory–practice gap. Addressing them will require a holistic view of system design and collaboration between ML, security and human-computer interaction communities. We treat this emerging area as a direction for future research rather than as part of our systematic evaluation.

11 Limitations and Threats to Validity

This study has several limitations. It focuses on four security conferences, so our observations may not reflect adversarial ML research in ML-focused venues or other domains where different norms prevail. Our coding and gap score simplify reality—binary flags cannot capture partial progress or nuanced threat models—and the classifications were done manually, which can introduce bias or overlook subtle details. The field is also evolving rapidly; our dataset ends in early 2025, and the balance of work may shift as research on large language models and other modalities expands. Finally, what counts as "practical" is context-dependent and may change over time: assumptions impractical today might become realistic tomorrow as technology and attacker capabilities evolve. Despite these caveats, we hope that our synthesis of quantitative trends and qualitative themes offers useful insights and prompts more systematic evaluations in future studies.

12 Conclusion

Over the 2022–2025 period our review of 454 adversarial ML papers shows that the field has grown in technical depth and has begun to engage more with practical considerations. About one study in twenty evaluated methods on a deployed system, and many continued to assume white-box access, gradient information or generous query budgets. These simplifications are understandable given the complexity of real deployments but highlight where future work can focus. In addition to quantitative gaps, we observed qualitative mismatches: common benchmarks emphasise L_p norms and worst-case analysis rather than factors more relevant to practitioners such as human perception, system integration or economic cost. Nevertheless, there are encouraging trends, including increased code release, emerging research on query-efficient and gradient-free methods, and a growing discourse on realistic threat

modelling, economic and human factors.

To align adversarial ML research with real-world challenges, the community should emphasise realistic threat models, limited attacker access, testing on real systems, diversification beyond vision benchmarks, and consideration of human and economic factors. Coordinated efforts across researchers, conferences, practitioners and funders can accelerate progress: validating methods under deployment-like conditions, valuing practical relevance in peer review, sharing data and evaluations, and supporting collaborative projects that bridge theory and practice. Ensuring that AI systems remain robust as they proliferate across high-stakes domains is a societal imperative, and we are optimistic that increased awareness and collaboration will help narrow the gap between adversarial ML research and operational security.

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A Complete Paper Analysis Dataset

For transparency and to facilitate future meta-analyses, we provide our full dataset of coded results for all 454 papers in the supplementary material (`all_conferences_analysis_results_2022_2025.csv`). Table 1 shows a condensed excerpt of key fields.

The dataset columns include:

- Year, Conference, Title (first author and shorthand)
- G1 (Focus: attack or defense), G2 (Attack Type: Evasion, Poison, Privacy, Multiple)
- G3 (ML paradigm: DL=deep learning, Trad=traditional ML, Both)
- G4 (Data Domain: Images, Text, Audio, Malware, Other)
- G5 (Economics discussed: YES/NO)
- G6 (Code released: YES/NO)
- G7 (Real system testing: YES/NO)
- T1 (Threat model: White-box, Gray-box, Black-box)
- T2 (Training data access assumed: Full, Partial, None)
- Q1 (Requires gradients: YES/NO)
- Q2 (Query budget: High, Low, None)
- Q3 (Computation needed: High = GPU or extensive, Low = CPU or minimal)
- Gap Score (0–6) as defined in the paper.

In the table, binary fields are indicated with checkmarks (Yes) or $\times$ (No), and abbreviated labels are used for threat model (W = white, B = black, G = gray) and other categorical fields for compactness.

Table 1: Summary of all reviewed papers with key taxonomy fields. (G1: Focus; G2: Attack Type; G3: ML Type; G4: Domain; G5: Economics; G6: Code; G7: Real-world eval; T1: Threat Model; T2: Training Data Access; Q1: Gradient; Q2: Query Budget; Q3: Computation; WB: White-box flag; Gap Score.)

Year	Venue	First Author	G1	G2	G3	G4	G5	G6	G7	T1	T2	Q1	Q2	Q3	WB	Gap
2022	ACM	Using	def	Po	DL	Img	✓	✓	×	G	F	✓	H	L	0	3
2022	ACM	Reverse-Engineering	atk	Ev	DL	Aud	✓	✓	×	B		×	H	L	0	2
2022	ACM	Tianshuo	def	Pr	DL	Img	✓	✓	×	B		×	L	L	0	1
2022	ACM	CISPA	atk	Pr	DL	Oth	✓	×	×	B		×	L	L	0	2
2022	ACM	CISPA	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	School	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Physical	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Framework	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Florian	atk	Po	DL	Img	✓	×	×	W	F	✓	H	L	1	5
2022	ACM	via	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	ExamplesforLeakage	def	Ev	DL	Img	✓	✓	×	W	P	✓	H	L	1	4
2022	ACM	Harnessing	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Feature	atk	Pr	DL	Oth	✓	×	✓	B		×	L	L	0	1
2022	ACM	Imperial	atk	Pr	Both	Oth	✓	✓	×	B		×	H	L	0	2
2022	ACM	Learning	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	CISPA	atk	Ev	DL	Txt	✓	✓	×	B		×	H	L	0	2
2022	ACM	Amrita	def	Po	DL	Img	✓	×	×	B		✓	H	L	0	4
2022	ACM	A	def	Ev	Both	Txt	✓	✓	×	B		×	L	L	0	1
2022	ACM	CISPA	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	ETH	def	Pr	Both	Oth	✓	×	×	B		×	L	L	0	2
2022	ACM	Hanqing	atk	Ev	DL	Aud	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Stevens	atk	Pr	DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Are	atk	Pr	DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	When	atk	Ev	DL	Aud	✓	✓	✓	B		×	H	H	0	1
2022	ACM	Learning	atk	Pr	DL	Img	✓	✓	×	B		×	H	L	0	2
2022	ACM	Using	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Neural	atk	Ev	DL	Txt	✓	✓	×	B	P	✓	H	H	0	3
2022	ACM	CISPA	atk	Pr	DL	Img	✓	✓	×	B	P	×	L	L	0	1
2022	ACM	A	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	LPGNet:	def	Pr	DL	Oth	✓	✓	×	B		✓	H	L	0	3
2022	IEEE	Encoders	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	1Illinois	atk	Ev	DL	Img	✓	✓	×	B		×	L	L	0	1
2022	IEEE	Graph	atk	Pr	DL	Oth	✓	✓	×	B	F	✓	H	L	0	3
2022	IEEE	Imperceptible	atk	Ev	DL	Txt	✓	✓	✓	B		×	L	L	0	0
2022	IEEE	Security	def		DL	Oth	✓	✓	×	B		×		L	0	1
2022	IEEE	Jialuo	def		DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Yun	atk	Ev	DL	Oth	✓	✓	×	B		×	H	L	0	2
2022	IEEE	Production	atk	Po	DL	Img	✓	×	×	W	P	✓	H	L	1	5
2022	IEEE	Classification	def		Both	Mal	✓	✓	×	B		×	L	L	0	1
2022	IEEE	in	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Markus	def	Pr	Both	Oth	✓	✓	×	B		×	L	L	0	1
2022	IEEE	Eugene	atk	Po	DL	Txt	✓	✓	×	B		✓	H	H	0	3
2022	IEEE	"Adversarial	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Attacks	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Weight	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	SoK:	def		Both	Oth	✓	×	×	B		×		L	0	2
2022	IEEE	Borja	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Transformer	atk	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Property	atk	Po	DL	Img	✓	✓	×	B	F	×	H	L	0	2
2022	IEEE	Yuhao	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2022	NDS	School	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	NDS	Federated	def	Po	DL	Txt	✓	×	×	W		×	L	L	1	3
2022	NDS	Property	atk	Pr	DL	Img	✓	✓	×	B	F	✓	H	L	0	3
2022	NDS	RamBoAttack:	atk	Ev	DL	Img	✓	✓	×	B		×	H	H	0	2
2022	NDS	Shengwei	atk	Pr	DL	Img	✓	✓	✓	W	F	✓	H	L	1	3
2022	NDS	Robustness	both	Po	DL	Img	✓	×	×	W	F	✓	H	L	1	5
2022	NDS	Ahmed	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	for	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	Vocal	def	Ev	DL	Aud	✓	✓	×	B		×		L	0	1
2022	USENIX	31st	atk	Pr	Both	Oth	✓	×	×	W	F	✓	H	L	1	5
2022	USENIX	August	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	sponsored	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4

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Year	Venue	First Author	G1	G2	G3	G4	G5	G6	G7	T1	T2	Q1	Q2	Q3	WB	Gap
2022	USENIX	August	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	sponsored	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	sponsored	atk	Po	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	Attacks	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	sponsored	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	August	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	August	atk	Pr	DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	Novel	atk	Pr	Both	Oth	✓	✓	×	B		×	H	L	0	2
2022	USENIX	Technische	def		Both	Mal	✓	×	×	B		×		L	0	2
2022	USENIX	1CISPA	atk	Pr	DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	CISPA	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Machine	def	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Virginia	atk	Po	DL	Img	✓	✓	×	G	P	✓	L	L	0	2
2023	ACM	through	atk	Pr	DL	Aud	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Stealing	atk	Ev	DL	Txt	✓	✓	×	B		×	H	L	0	2
2023	ACM	Stolen	both	M	DL	Oth	✓	✓	×	B	P	✓	H	L	0	3
2023	ACM	Information	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Deep	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Unforgeability	def		DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Stefano	def	Ev	Traditional	Img	✓	✓	×	B		×	L	L	0	1
2023	ACM	Learning	atk	Po	DL	Oth	✓	✓	×	G	P	✓	H	L	0	3
2023	ACM	Secure	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Detection	atk	Ev	DL	Mal	✓	✓	×	B		×	L	L	0	1
2023	ACM	Code	def		DL	Oth	✓	✓	×	G		×	L	L	0	1
2023	ACM	and	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	through	atk	Pr	DL	Oth	✓	✓	×	B		✓	H	L	0	3
2023	ACM	Jingxuan	def		DL	Oth	✓	✓	×	B		×	L	L	0	1
2023	ACM	and	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Evading	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Changqing	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	AntiFake:	def	Ev	DL	Aud	✓	×	✓	B		×	H	L	0	2
2023	ACM	against	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Effects	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Deep	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	arXiv:2112.04558v2	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Federated	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	SNAP:	atk	Po	DL	Oth	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Federated	def	Po	DL	Img	✓	×	×	W	F	✓	H	L	1	5
2023	IEEE	Information	atk	Pr	DL	Txt	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Covert	atk	Po	DL	Img	✓	✓	×	B		✓	H	L	0	3
2023	IEEE	DepthFake	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Jiameng	atk	Ev	DL	Txt	✓	✓	×	B		×		L	0	1
2023	IEEE	Multi-party	def	Pr	Both	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Andrew	def	Pr	Both	Oth	✓	×	×	W	F	✓	H	L	1	5
2023	IEEE	Information	atk	Pr	DL	Txt	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Extraction	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Limin	atk	Po	DL	Mal	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Learning	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Attacks	both	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Automatic	atk	Ev	DL	Aud	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	yCybersecurity	def		DL	Img	✓	×	×	W		×	L	L	1	3
2023	IEEE	Ruijie	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	A	both	Pr	DL	Img	✓	✓	×	B	P	×	H	L	0	2
2023	IEEE	SNAP:	atk	Po	DL	Oth	✓	×	×	B		×	H	L	0	3
2023	IEEE	Hong	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	BayBFed	def	Po	DL	Img	✓	×	×	W	F	✓	H	L	1	5
2023	IEEE	Andre	atk	Ev	DL	Aud	✓	✓	×	B	F	×	L	L	0	1
2023	IEEE	KASTEL	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	ELSA:	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Shengwei	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Patch-agnostic	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	RAB:	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	ETH	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	and	def	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	through	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	StyleFool:	atk	Ev	DL	Img	✓	✓	×	B		×	L	L	0	1
2023	NDS	DNN	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Hadi	def	Ev	DL	Aud	✓	✓	×	B		×	H	L	0	2

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Year	Venue	First Author	G1	G2	G3	G4	G5	G6	G7	T1	T2	Q1	Q2	Q3	WB	Gap
2023	NDS	Yugeng	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Commercial	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Classifiers	def	Ev	DL	Img	✓	✓	×	B		×	L	L	0	1
2023	NDS	Learning	def	Ev	DL	Oth	✓	✓	×	B		✓	H	L	0	3
2023	NDS	Jiayun	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Klim	atk	Ev	DL	Oth	✓	×	×	B		×	H	H	0	3
2023	NDS	Siyuan	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Universidad	def	Po	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Deep	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Chunyi	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Robust	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Physical	def	Ev	Both	Oth	✓	✓	✓	G	P	×	L	L	0	0
2023	NDS	*Technische	def	Pr	Both	Txt	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX		atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Detection	atk	Ev	DL	Mal	✓	×	×	B		×	H	L	0	3
2023	USENIX	Scientific	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	of	def		DL	Oth	✓	✓	×	B		×	L	L	0	1
2023	USENIX	for	atk	Po	Both	Oth	✓	×	×	W	F	✓	H	L	1	5
2023	USENIX	via	def	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	FREE	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	1CISPA	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Jonathan	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Vehicles	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Fairness	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Fairness	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Decompiling	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	August	def		Both	Mal	✓	✓	×	B		×	L	L	0	1
2023	USENIX	Mazal	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	August	def		DL	Mal	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	is	atk	Pr	DL	Img	✓	✓	×	B	P	✓	H	L	0	3
2023	USENIX	is	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Federated	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Kunal	atk	Ev	DL	Mal	✓	✓	×	B	P	×	H	L	0	2
2023	USENIX	for	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	by	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	is	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	is	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	August	atk	Po	DL	Txt	✓	×	×	B		×	H	L	0	3
2023	USENIX	Universal	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2023	USENIX	is	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	August	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	from	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Xueluan	def	Pr	DL	Img	✓	×	×	B		×	L	L	0	2
2024	ACM	SafeEar:	def	Pr	DL	Aud	✓	✓	×	B		×	L	L	0	1
2024	ACM	Against	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Multi-Agent	atk	Ev	DL	Img	✓	✓	×	B		×	H	H	0	2
2024	ACM	Text-to-Image	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Autonomous	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Wei	atk	Po	Both	Oth	✓	✓	×	G	P	✓	H	L	0	3
2024	ACM	Byzantine-Robust	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Recognition	atk	Ev	DL	Aud	✓	✓	✓	B		×		H	0	0
2024	ACM	Zijin	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	CSIRO's	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Items	atk	Po	DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Applications	atk	Pr	DL	Txt	✓	✓	×	B		✓	H	H	0	3
2024	ACM	Attack	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Attachment	atk	Ev	DL	Mal	✓	✓	✓	B		×	H	L	0	1
2024	ACM	Information	atk	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Alchemy:	def	Ev	DL	Img	✓	✓	×	W		✓	H	L	1	4
2024	ACM	S2NeRF:	both	Pr	DL	Img	✓	✓	×	B		✓	H	L	0	3
2024	ACM	Certified	atk	Po	DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	ETH	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Jie	atk	Pr	DL	Img	✓	✓	×	B	P	×	H	L	0	2
2024	ACM	Understanding	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	in	def	Ev	DL	Mal	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	A	def	Pr	DL	Img	✓	✓	×	B		×	H	L	0	2

(Continued on next page)

Year	Venue	First Author	G1	G2	G3	G4	G5	G6	G7	T1	T2	Q1	Q2	Q3	WB	Gap
2024	ACM	The	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Fisher	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	into	atk	Pr	DL	Txt	✓	✓	×	B		×	H	L	0	2
2024	ACM	BadMerging:	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Optimization-based	atk	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Optimization-based	atk	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Model	atk	Pr	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Shenzhen	atk	Pr	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Information	def	Ev	DL	Img	✓	✓	×	B		×		H	0	1
2024	ACM	Learning	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Blind	def		Both	Aud	×	×	×	B		×		L	0	3
2024	ACM	Information	atk	Pr	DL	Txt	✓	×	×	B		×	L	L	0	2
2024	ACM	Deepfake	def		DL	Aud	✓	✓	×	B		×		L	0	1
2024	ACM	Membership	atk	Pr	DL	Img	✓	✓	×	B	P	×	L	L	0	1
2024	ACM	Membership	atk	Pr	DL	Img	✓	✓	×	B	P	×	L	L	0	1
2024	ACM	Neural	atk	Pr	DL	Img	✓	✓	×	B		×		L	0	1
2024	ACM		atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Fine-grained	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	from	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Institute	atk	Pr	DL	Img	✓	✓	×	B		×	H	L	0	2
2024	ACM	Institute	atk	Pr	DL	Img	✓	✓	×	B	F	×	H	L	0	2
2024	ACM	of	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	School	atk	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Examples:	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2024	ACM	Models	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Learning	atk	Po	DL	Img	✓	✓	×	B	P	×	L	L	0	1
2024	IEEE	Pattern	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Robin	def	Pr	Both	Oth	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Yuzheng	def	Pr	Both	Img	✓	✓	×	B		×	L	L	0	1
2024	IEEE	Joshua	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Ziqi	def	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Linguistic	atk	Ev	DL	Aud	✓	✓	✓	B		×	L	L	0	0
2024	IEEE	Language	atk	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	It's	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	into	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2024	IEEE	Test-Time	atk	Po	DL	Img	✓	✓	×	G	P	✓	L	L	0	2
2024	IEEE	Xingshuo	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Mohammad	atk	Po	DL	Img	✓	×	×	G	P	✓	H	L	0	4
2024	IEEE	Attacks	def	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2024	IEEE	Distribution	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Dropout	atk	Po	DL	Img	✓	×	×	W	F	✓	H	L	1	5
2024	IEEE	GROV	def	Ev	DL	Oth	✓	✓	×	B		×	L	L	0	1
2024	IEEE	Mahmoud	atk	Ev	DL	Oth	✓	✓	×	B	P	×	L	L	0	1
2024	IEEE	Alec	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Accelerator	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	KASTEL	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Adversarial	def	Ev	DL	Txt	×	✓	×	W		×		L	1	3
2024	IEEE	2Services	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Large	def	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	DEMASQ:	def	Ev	DL	Txt	✓	×	×	B		✓	H	L	0	4
2024	NDS	Attacks	atk	Pr	DL	Aud	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Poisoning	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	by	def	Pr	DL	Img	✓	✓	×	B		×	L	L	0	1
2024	NDS	CamPro	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Ensemble	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Gelei	atk	Ev	DL	Txt	✓	×	×	B		×	H	H	0	3
2024	NDS	Watermarking	def	Ev	DL	Aud	✓	✓	✓	W	F	×	L	L	1	1
2024	NDS	ICS	atk	Ev	DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Federated	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Chengkun	def	Po	DL	Txt	✓	✓	×	B		✓	L	L	0	2
2024	NDS	Recognition	atk	Ev	DL	Aud	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Transpose	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Transpose	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Inversion-based	def	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Networks	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4

(Continued on next page)

Year	Venue	First Author	G1	G2	G3	G4	G5	G6	G7	T1	T2	Q1	Q2	Q3	WB	Gap
2024	NDS	Browser	def	Pr	Traditional	Oth	✓	×	×	B		×		L	0	2
2024	NDS	Deep	def	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2024	NDS	Bang	def	Pr	DL	Oth	✓	✓	×	B		×	L	L	0	1
2024	NDS	Against	atk	Po	DL	Img	✓	✓	×	B	P	✓	H	L	0	3
2024	NDS	Speaker	atk	Ev	DL	Aud	✓	✓	✓	B		×		L	0	0
2024	NDS	Mitigating	def	Po	DL	Img	✓	×	×	B		×		L	0	2
2024	NDS	for	def		DL	Mal	✓	✓	×	B		×	L	L	0	1
2024	NDS	in	both	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Attacks	def	Po	DL	Txt	✓	✓	×	W	F	×	L	L	1	2
2024	NDS	Chaoxiang	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Tong	atk	Ev	DL	Txt	✓	✓	×	B		×	L	L	0	1
2024	USENIX	Attacks	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Approach	def	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Qingzhao	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Zhenghang	def		Both	Oth	✓	✓	×	B		×		L	0	1
2024	USENIX	Synthesis	def	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Measuring	def		DL	Aud	✓	✓	×	B		×	L	L	0	1
2024	USENIX	Large	atk	Ev	DL	Txt	✓	✓	×	B		×	H	L	0	2
2024	USENIX	Efficiency	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Efficiency	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Kathrin	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	DNN-GP:	both	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	for	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	DeepEclipse	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	DeepEclipse	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Between	atk	M	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Red-Teaming	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2024	USENIX	Red-Teaming	atk	Ev	DL	Img	✓	✓	×	B		×	H	L	0	2
2024	USENIX	Property	atk	Pr	DL	Img	✓	✓	×	B	F	×	H	L	0	2
2024	USENIX	Transposed	def		DL	Img	✓	×	×	W	F	✓	H	L	1	5
2024	USENIX	in	def	Pr	DL	Img	✓	✓	×	B		×	L	L	0	1
2024	USENIX	Shuaifan	def	Pr	DL	Img	✓	✓	✓	W	F	✓	H	L	1	3
2024	USENIX	CISPA	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	CISPA	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Runtime	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Splitting	def	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Yixin	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Changjiang	atk	Po	DL	Img	✓	✓	×	B	F	✓	H	L	0	3
2024	USENIX	Guangsheng	atk	Pr	DL	Img	✓	×	×	W	F	✓	H	L	1	5
2024	USENIX	Bailey	def	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Meng	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	"Minxue	def	Pr	DL	Img	✓	✓	×	W		✓	H	L	1	4
2024	USENIX	CISPA	atk	Pr	DL	Img	✓	✓	×	B	P	×	H	H	0	2
2024	USENIX	False	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Shenchen	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	through	atk	Pr	DL	Oth	✓	×	×	B		✓	H	L	0	4
2024	USENIX	Shaofeng	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Mudjacking:	def	Po	DL	Img	✓	×	×	B		✓	H	L	0	4
2024	USENIX	Randomised	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	33rd	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	is	def		DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	August	def	Pr	Both	Txt	✓	✓	×	B		×	L	L	0	1
2024	USENIX	August	atk	Pr	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	is	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	is	def		DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	is	def	Po	DL	Img	✓	×	×	B		×		L	0	2
2024	USENIX	Attacks	def	Pr	DL	Img	✓	✓	×	B		✓	H	L	0	3
2024	USENIX	Model	atk	Ev	DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Zooming	both	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	August	atk	Pr	DL	Txt	✓	✓	×	B		×	L	L	0	1
2024	USENIX	is	def		DL	Txt	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	August	def		DL	Oth	✓	✓	×	W	F	×	H	L	1	3
2024	USENIX	August	def		DL	Oth	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	August	def	Ev	DL	Img	✓	✓	×	B		×		L	0	1
2024	USENIX	August	def		Both	Mal	✓	×	×	B		×	L	L	0	2
2024	USENIX	Learning	def	Ev	Both	Oth	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Framework	def		DL	Txt	✓	✓	×	B		×		L	0	1
2024	USENIX		def		DL	Img	✓	✓	×	B		×		L	0	1

(Continued on next page)

Year	Venue	First Author	G1	G2	G3	G4	G5	G6	G7	T1	T2	Q1	Q2	Q3	WB	Gap
2024	USENIX		atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Inference	def	Pr	DL	Img	✓	×	×	G		✓	H	L	0	4
2025	ACM	1Technion	atk	Ev	DL	Txt	✓	✓	×	B		×	H	L	0	2
2025	ACM	in	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	arXiv:2501.05928v2	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Anti-Facial	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	BIFOLD	atk	Ev	DL	Oth	✓	✓	×	W	F	✓	H	H	1	4
2025	ACM	Targeted	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Language	atk	Po	DL	Txt	✓	✓	×	B		×	L	L	0	1
2025	ACM	Models	def	Ev	DL	Img	✓	×	✓	G	P	✓	H	H	0	3
2025	ACM	Adversarial	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Construction	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Latent-based	atk	Ev	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Milad	atk	Ev	DL	Mal	✓	✓	✓	W	F	✓	H	L	1	3
2025	ACM	VillainNet:	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Deep	def		DL	Mal	✓	✓	×	B		×		L	0	1
2025	USENIX	Jiachen	def	Ev	DL	Img	✓	✓	✓	B		✓	H	H	0	2
2025	USENIX	Lingchen	def	Ev	DL	Txt	✓	✓	×	W	F	×	L	L	1	2
2025	USENIX	Revisiting	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	USENIX	Zhisheng	def	Pr	DL	Aud	✓	✓	×	B		×	L	L	0	1
2025	USENIX	with	def	Ev	DL	Oth	✓	✓	×	W	F	×	H	L	1	3
2025	USENIX	Persistent	atk	Po	DL	Img	✓	✓	×	W		✓	H	L	1	4
2025	USENIX	Benign	atk	Po	DL	Img	✓	✓	×	W	F	✓	H	L	1	4
2025	USENIX	34th	def	Ev	DL	Mal	✓	✓	×	B		✓	H	L	0	3